**PHYS 210 Electricity and Magnetism**

**Celebration of Knowledge!**

Supplementary Practice 1 - CK #2 preparation

Name: _____ Date: _____

Time: _____ Marks: _____ Practice paper - not an official exam

Attempt all questions. Supporting practice; this topic mix is not the announced main-test format.

Write all rough calculations and final answers on separate paper. Show your working.

Original source numbering is retained inside each question. Answer the entire cropped question.

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**1. [____]** Tutorial 4 - Question 4

4. A long nonconducting cylinder (radius = 6.0 mm) has a nonuniform volume charge density given by $\alpha r^2$, where $\alpha = 6.2\ \text{mC/m}^5$ and $r$ is the distance from the axis of the cylinder. What is the magnitude of the electric field at a point 2.0 mm from the axis?
   a. 1.4 N/C
   b. 1.6 N/C
   c. 1.8 N/C
   d. 2.0 N/C
   e. 5.4 N/C

**2. [____]** Homework 5 - Question 23.1

**23.1** •• A point charge $q_1 = +2.40\ \mu\text{C}$ is held stationary at the origin. A second point charge $q_2 = -4.30\ \mu\text{C}$ moves from the point $x = 0.150$ m, $y = 0$ to the point $x = 0.250$ m, $y = 0.250$ m. How much work is done by the electric force on $q_2$?

**3. [____]** Homework 6 - Question 24.1

**24.1** • The plates of a parallel-plate capacitor are 2.50 mm apart, and each carries a charge of magnitude 80.0 nC. The plates are in vacuum. The electric field between the plates has a magnitude of $4.00 \times 10^6$ V/m. What is (a) the potential difference between the plates; (b) the area of each plate; (c) the capacitance?

**4. [____]** Homework 6 - Question 24.3

**24.3** • A parallel-plate air capacitor of capacitance 245 pF has a charge of magnitude 0.148 $\mu$C on each plate. The plates are 0.328 mm apart. (a) What is the potential difference between the plates? (b) What is the area of each plate? (c) What is the electric-field magnitude between the plates? (d) What is the surface charge density on each plate?

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Supplementary Practice 2 - CK #2 preparation

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**1. [____]** Homework 6 & 7 - Question 24.7

**24.7** • A parallel-plate air capacitor is to store charge of magnitude 240.0 pC on each plate when the potential difference between the plates is 42.0 V. (a) If the area of each plate is 6.80 $cm^2$, what is the separation between the plates? (b) If the separation between the two plates is double the value calculated in part (a), what potential difference is required for the capacitor to store charge of magnitude 240.0 pC on each plate?

**2. [____]** Homework 7 - Question 24.8

**24.8** • A 5.00-pF, parallel-plate, air-filled capacitor with circular plates is to be used in a circuit in which it will be subjected to potentials of up to $1.00 \times 10^2$ V. The electric field between the plates is to be no greater than $1.00 \times 10^4$ N/C. As a budding electrical engineer for Live-Wire Electronics, your tasks are to (a) design the capacitor by finding what its physical dimensions and separation must be; (b) find the maximum charge these plates can hold.

**3. [____]** Homework 7 - Question 24.24

**24.24** • A parallel-plate air capacitor has a capacitance of 920 pF. The charge on each plate is 3.90 $\mu$C. (a) What is the potential difference between the plates? (b) If the charge is kept constant, what will be the potential difference if the plate separation is doubled? (c) How much work is required to double the separation?

**4. [____]** Homework 7 - Question 24.36

**24.36** • A budding electronics hobbyist wants to make a simple 1.0-nF capacitor for tuning her crystal radio, using two sheets of aluminum foil as plates, with a few sheets of paper between them as a dielectric. The paper has a dielectric constant of 3.0, and the thickness of one sheet of it is 0.20 mm. (a) If the sheets of paper measure 22 × 28 cm and she cuts the aluminum foil to the same dimensions, how many sheets of paper should she use between her plates to get the proper capacitance? (b) Suppose for convenience she wants to use a single sheet of posterboard, with the same dielectric constant but a thickness of 12.0 mm, instead of the paper. What area of aluminum foil will she need for her plates to get her 1.0 nF of capacitance? (c) Suppose she goes high-tech and finds a sheet of Teflon of the same thickness as the posterboard to use as a dielectric. Will she need a larger or smaller area of Teflon than of posterboard? Explain.

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Supplementary Practice 3 - CK #2 preparation

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**1. [____]** Tutorial 6 - Question 7

7. An air-filled capacitor consists of two parallel plates, each with an area of 7.60 $cm^2$, separated by a distance of 1.80 mm. A 20.0-V potential difference is applied to these plates. Calculate (a) the electric field between the plates, (b) the surface charge density, (c) the capacitance, and (d) the charge on each plate.

**2. [____]** Tutorial 6 - Question 31

**31.** (a) A 3.00-$\mu$F capacitor is connected to a 12.0-V battery. How much energy is stored in the capacitor? (b) If the capacitor had been connected to a 6.00-V battery, how much energy would have been stored?

**3. [____]** Tutorial 6 - Question 34

**34.** A parallel-plate capacitor is charged and then disconnected from a battery. By what fraction does the stored energy change (increase or decrease) when the plate separation is doubled?

**4. [____]** Tutorial 7 - Question 6

6. A parallel plate capacitor of capacitance $C_0$ has plates of area $A$ with separation $d$ between them. When it is connected to a battery of voltage $V_0$, it has charge of magnitude $Q_0$ on its plates. The plates are pulled apart to a separation $2d$ while the capacitor remains connected to the battery. After the plates are $2d$ apart, the capacitance of the capacitor and the magnitude of the charge on the plates are
   a. $\frac{1}{2}C_0, \frac{1}{2}Q_0$
   b. $\frac{1}{2}C_0, Q_0$
   c. $C_0, Q_0$
   d. $2C_0, Q_0$
   e. $2C_0, 2Q_0$

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Supplementary Practice 4 - CK #2 preparation

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**1. [____]** Textbook 15e - Question VP24.11.2

**VP24.11.2** The two plates of a parallel-plate capacitor each have area 1.20 $m^2$, are 4.50 mm apart, and initially have vacuum between them. You connect a power supply to the capacitor, charging it to 3.50 kV. Keeping the power supply connected, you insert a sheet of dielectric with dielectric constant 2.50 that fills the space between the plates.

Find (a) the original capacitance, (b) the original charge on the capacitor plates, (c) the new capacitance once you insert the dielectric, (d) the new charge on the capacitor plates, and (e) the magnitude of the induced charge on the faces of the dielectric sheet.

**2. [____]** Textbook 15e - Question VP24.11.3

**VP24.11.3** With vacuum between its plates, a parallel-plate capacitor has capacitance 4.50 $\mu$F. You attach a power supply to the capacitor, charging it to 2.80 kV, and then disconnect it. You then insert a dielectric sheet that fills the space between the plates. The potential difference between the plates decreases to 1.20 kV, and the charge on each plate remains constant. Find the energy stored in the capacitor (a) before you insert the sheet and (b) after you insert the sheet.

**3. [____]** Textbook 15e - Question VP24.11.4

**VP24.11.4** With vacuum between its plates, a parallel-plate capacitor has capacitance 5.60 $\mu$F. You attach a power supply to the capacitor, charging it to 1.50 kV. Keeping the power supply connected, you insert a dielectric sheet with dielectric constant 2.85 that fills the space between the plates. Find the energy stored in the capacitor (a) before the sheet is inserted and (b) after the sheet is inserted.

**4. [____]** Homework 4 - Question 22.15

**22.15** •• How many excess electrons must be added to an isolated spherical conductor 26.0 cm in diameter to produce an electric field of magnitude 1150 N/C just outside the surface?

**PHYS 210 Electricity and Magnetism**

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Supplementary Practice 5 - CK #2 preparation

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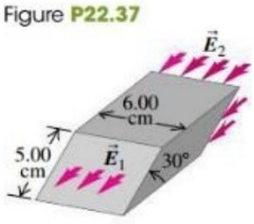
Original source numbering is retained inside each question. Answer the entire cropped question.

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**1. [____]** Homework 4 - Question 22.37

**22.37** •• The electric field $\vec{E}_1$ at one face of a parallelepiped is uniform over the entire face and is directed out of the face. At the opposite face, the electric field $\vec{E}_2$ is also uniform over the entire face and is directed into that face (**Fig. P22.37**). The two faces in question are inclined at 30.0° from the horizontal, while both $\vec{E}_1$ and $\vec{E}_2$ are horizontal; $\vec{E}_1$ has a magnitude of $2.50 \times 10^4$ N/C, and $\vec{E}_2$ has a magnitude of $7.00 \times 10^4$ N/C. (a) Assuming that no other electric field lines cross the surfaces of the parallelepiped, determine the net charge contained within. (b) Is the electric field produced by the charges only within the parallelepiped, or is the field also due to charges outside the parallelepiped? How can you tell?

Figure P22.37




**2. [____]** Tutorial 4 - Question 1

1. Two infinite, uniformly charged, flat surfaces are mutually perpendicular. One of the sheets has a charge density of +60 pC/m$^2$, and the other carries a charge density of –80 pC/m$^2$. What is the magnitude of the electric field at any point not on either surface?
   a. 1.1 N/C
   b. 5.6 N/C
   c. 7.9 N/C
   d. 3.8 N/C
   e. 4.0 N/C

**3. [____]** Tutorial 4 - Question 2

2. A long nonconducting cylinder (radius = 12 cm) has a charge of uniform density (5.0 nC/m$^3$) distributed throughout its column. Determine the magnitude of the electric field 5.0 cm from the axis of the cylinder.
   a. 25 N/C
   b. 20 N/C
   c. 14 N/C
   d. 31 N/C
   e. 34 N/C

**4. [____]** Tutorial 4 - Question 3

3. Each 2.0-m length of a long cylinder (radius = 4.0 mm) has a charge of 4.0 nC distributed uniformly throughout its volume. What is the magnitude of the electric field at a point 5.0 mm from the axis of the cylinder?
   a. 9.9 kN/C
   b. 8.1 kN/C
   c. 9.0 kN/C
   d. 7.2 kN/C
   e. 18 kN/C

**PHYS 210 Electricity and Magnetism**

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Supplementary Practice 6 - CK #2 preparation

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**1. [____]** Tutorial 4 - Question 5

5. Charge of uniform density (80 nC/m$^3$) is distributed throughout a hollow cylindrical region formed by two coaxial cylindrical surfaces of radii 1.0 mm and 3.0 mm. Determine the magnitude of the electric field at a point which is 2.0 mm from the symmetry axis.
   a. 7.9 N/C
   b. 9.0 N/C
   c. 5.9 N/C
   d. 6.8 N/C
   e. 18 N/C

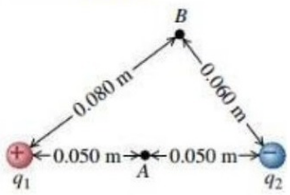
**2. [____]** Homework 5 - Question 23.8

**23.8** •• Three equal 1.20-$\mu$C point charges are placed at the corners of an equilateral triangle with sides 0.400 m long. What is the potential energy of the system? (Take as zero the potential energy of the three charges when they are infinitely far apart.)

**3. [____]** Homework 5 - Question 23.19

**23.19** • Two point charges $q_1 = +2.40$ nC and $q_2 = -6.50$ nC are 0.100 m apart. Point $A$ is midway between them; point $B$ is 0.080 m from $q_1$ and 0.060 m from $q_2$ (**Fig. E23.19**). Take the electric potential to be zero at infinity. Find (a) the potential at point $A$; (b) the potential at point $B$; (c) the work done by the electric field on a charge of 2.50 nC that travels from point $B$ to point $A$.

Figure **E23.19**




**4. [____]** Homework 5 - Question 23.27

**23.27** •• A uniformly charged, thin ring has radius 15.0 cm and total charge +24.0 nC. An electron is placed on the ring's axis a distance 30.0 cm from the center of the ring and is constrained to stay on the axis of the ring. The electron is then released from rest. (a) Describe the subsequent motion of the electron. (b) Find the speed of the electron when it reaches the center of the ring.

**PHYS 210 Electricity and Magnetism**

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Supplementary Practice 7 - CK #2 preparation

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**1. [____]** Homework 6 - Question 23.44

**23.44** • CALC In a certain region of space the electric potential is given by $V = +Ax^2y - Bxy^2$, where $A = 5.00\ \text{V/m}^3$ and $B = 8.00\ \text{V/m}^3$. Calculate the magnitude and direction of the electric field at the point in the region that has coordinates $x = 2.00$ m, $y = 0.400$ m, and $z = 0$.

**2. [____]** Homework 6 - Question 23.55

**23.55** •• CALC A vacuum tube diode consists of concentric cylindrical electrodes, the negative cathode and the positive anode. Because of the accumulation of charge near the cathode, the electric potential between the electrodes is given by

$$V(x) = Cx^{4/3}$$

where $x$ is the distance from the cathode and $C$ is a constant, characteristic of a particular diode and operating conditions. Assume that the distance between the cathode and anode is 13.0 mm and the potential difference between electrodes is 240 V. (a) Determine the value of $C$. (b) Obtain a formula for the electric field between the electrodes as a function of $x$. (c) Determine the force on an electron when the electron is halfway between the electrodes.

**3. [____]** Homework 7 - Question 24.12

**24.12** •• A cylindrical capacitor has an inner conductor of radius 2.2 mm and an outer conductor of radius 3.5 mm. The two conductors are separated by vacuum, and the entire capacitor is 2.8 m long. (a) What is the capacitance per unit length? (b) The potential of the inner conductor is 350 mV higher than that of the outer conductor. Find the charge (magnitude and sign) on both conductors.

**4. [____]** Tutorial 5 - Question 3

**3.** (a) Calculate the speed of a proton that is accelerated from rest through a potential difference of 120 V. (b) Calculate the speed of an electron that is accelerated through the same potential difference.

**PHYS 210 Electricity and Magnetism**

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Supplementary Practice 8 - CK #2 preparation

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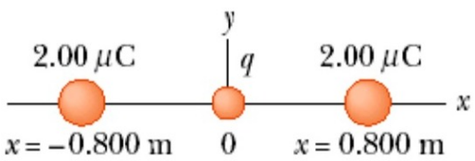
**1. [____]** Tutorial 5 - Question 37

**37.** The potential in a region between $x = 0$ and $x = 6.00$ m is $V = a + bx$, where $a = 10.0$ V and $b = -7.00$ V/m. Determine (a) the potential at $x = 0$, 3.00 m, and 6.00 m, and (b) the magnitude and direction of the electric field at $x = 0$, 3.00 m, and 6.00 m.

*Note:* Unless stated otherwise, assume the reference level of potential is $V = 0$ at $r = \infty$.

**2. [____]** Tutorial 5 - Question 16

**16.** Given two 2.00-$\mu$C charges, as shown in the figure below, and a positive test charge $q = 1.28 \times 10^{-18}$ C at the origin, (a) what is the net force exerted by the two 2.00-$\mu$C charges on the test charge $q$? (b) What is the electric field at the origin due to the two 2.00-$\mu$C charges? (c) What is the electrical potential at the origin due to the two 2.00-$\mu$C charges?




**3. [____]** Tutorial 5 - Question 17

**17.** At a certain distance from a point charge, the magnitude of the electric field is 500 V/m and the electric potential is –3.00 kV. (a) What is the distance to the charge? (b) What is the magnitude of the charge?

**4. [____]** Tutorial 5 - Question 18

**18.** A charge $+q$ is at the origin. A charge $-2q$ is at $x = 2.00$ m on the $x$ axis. For what finite value(s) of $x$ is (a) the electric field zero? (b) the electric potential zero?

**PHYS 210 Electricity and Magnetism**

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Supplementary Practice 9 - CK #2 preparation

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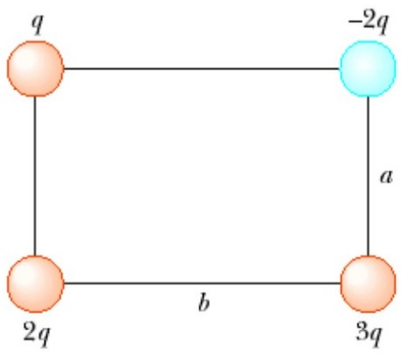
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**1. [____]** Tutorial 5 - Question 34

**34.** Calculate the energy required to assemble the array of charges shown in the figure below, where $a = 0.200$ m, $b = 0.400$ m, and $q = 6.00\ \mu$C.




___

**2. [____]** Tutorial 5 - Question 39

**39.** Over a certain region of space, the electric potential is $V = 5x - 3x^2y + 2yz^2$. Find the expressions for the $x$, $y$, and $z$ components of the electric field over this region. What is the magnitude of the field at the point $P$ that has coordinates (1, 0, –2) m?

**3. [____]** Tutorial 5 - Question 42

**42.** Consider a ring of radius $R$ with the total charge $Q$ spread uniformly over its perimeter. What is the potential difference between the point at the center of the ring and a point on its axis a distance $2R$ from the center?

___

**4. [____]** Tutorial 5 - Question 63

**63.** From Gauss's law, the electric field set up by a uniform line of charge is

$$E = \lambda / 2\pi\varepsilon_0 r$$

where $\lambda$ is the linear charge density along the line. Derive an expression for the potential difference between $r = r_1$ and $r = r_2$.

**PHYS 210 Electricity and Magnetism**

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Supplementary Practice 10 - CK #2 preparation

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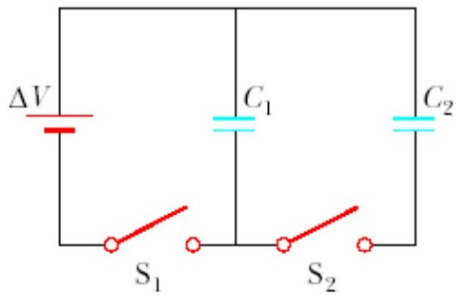
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**1. [____]** Tutorial 6 - Question 23

**23.** Consider the circuit shown in Figure P26.23, where $C_1 = 6.00\ \mu F$, $C_2 = 3.00\ \mu F$, and $\Delta V = 20.0$ V. Capacitor $C_1$ is first charged by the closing of switch $S_1$. Switch $S_1$ is then opened, and the charged capacitor is connected to the uncharged capacitor by the closing of $S_2$. Calculate the initial charge acquired by $C_1$ and the final charge on each capacitor.




**2. [____]** Tutorial 6 - Question 67

**67.** An isolated capacitor of unknown capacitance has been charged to a potential difference of 100 V. When the charged capacitor is then connected in parallel to an uncharged 10.0-$\mu$F capacitor, the potential difference across the combination is 30.0 V. Calculate the unknown capacitance.

**3. [____]** Tutorial 6 - Question 71

**71.** Capacitors $C_1 = 6.00\ \mu F$ and $C_2 = 2.00\ \mu F$ are charged as a parallel combination across a 250-V battery. The capacitors are disconnected from the battery and from each other. They are then connected positive plate to negative plate and negative plate to positive plate. Calculate the resulting charge on each capacitor.

**4. [____]** Tutorial 7 - Question 3

3. A 6.0-$\mu$F capacitor charged to 50 V and a 4.0-$\mu$F capacitor charged to 34 V are connected to each other, with the two positive plates connected and the two negative plates connected. What is the total energy stored in the 6.0-$\mu$F capacitor at equilibrium?
   a. 6.1 mJ
   b. 5.7 mJ
   c. 6.6 mJ
   d. 7.0 mJ
   e. 3.8 mJ

**PHYS 210 Electricity and Magnetism**

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Supplementary Practice 11 - CK #2 preparation

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Time: _____ Marks: _____ Practice paper - not an official exam

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**1. [____]** Tutorial 7 - Question 4

4. A 25-$\mu$F capacitor charged to 50 V and a capacitor $C$ charged to 20 V are connected to each other, with the two positive plates connected and the two negative plates connected. The final potential difference across the 25-$\mu$F capacitor is 36 V. What is the value of the capacitance of $C$?
   a. 43 $\mu$F
   b. 29 $\mu$F
   c. 22 $\mu$F
   d. 58 $\mu$F
   e. 63 $\mu$F

**2. [____]** Tutorial 7 - Question 5

5. A 30-$\mu$F capacitor is charged to 80 V and then connected across an initially uncharged capacitor of unknown capacitance $C$. If the final potential difference across the 30-$\mu$F capacitor is 20 V, determine $C$.
   a. 60 $\mu$F
   b. 75 $\mu$F
   c. 45 $\mu$F
   d. 90 $\mu$F
   e. 24 $\mu$F

**3. [____]** Textbook 15e - Question 24.49

**24.49** •• A 20.0 $\mu$F capacitor is charged to a potential difference of 800 V. The terminals of the charged capacitor are then connected to those of an uncharged 10.0 $\mu$F capacitor. Compute (a) the original charge of the system, (b) the final potential difference across each capacitor, (c) the final energy of the system, and (d) the decrease in energy when the capacitors are connected.

**4. [____]** Textbook 15e - Question 24.57

**24.57** •• Three capacitors having capacitances of 8.4, 8.4, and 4.2 $\mu$F are connected in series across a 36 V potential difference. (a) What is the charge on the 4.2 $\mu$F capacitor? (b) What is the total energy stored in all three capacitors? (c) The capacitors are disconnected from the potential difference without allowing them to discharge. They are then reconnected in parallel with each other, with the positively charged plates connected together. What is the voltage across each capacitor in the parallel combination? (d) What is the total energy now stored in the capacitors?